Near-Inertial Rotation in Compactly Supported Covariances: Exact Gaussian Processes on Half a Million Ocean Drifter Velocities

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Abstract

Surface drifters released by the hundreds sample ocean currents at submesoscale resolution, but mapping half a million drifter velocities with calibrated uncertainty has so far required approximate Gaussian processes (GPs). We fit exact GPs to all 1,079,888 scalar velocity observations of the LASER experiment on 32 CPU cores, using compactly supported matrix-valued kernels whose covariance matrices are sparse enough for supernodal Cholesky factorisation with exact log-determinants. Our pre-registered kernel—divergence- and curl-free terms derived from Wendland potentials plus a rotary near-inertial term, all on a 5-km support—only tied the scalar gp2Scale Wendland kernel on 48 held-out drifters. Diagnostics of the training residuals show why: their dominant structured component is a clockwise near-inertial oscillation coherent over at least 30 km. Carrying it explicitly resolves the tie. A local harmonic mean improves both kernels by 15% in RMSE; a rotary covariance on a 30-km, 12-h ellipsoid, affordable because the Cholesky factor of clustered drifter data grows only in proportion to its sparsity pattern, gives the best model. Its learned rotation frequency stays within 10% of the inertial frequency, it beats a two-range Wendland kernel with identical support by 3.5% in RMSE and 7.3% in CRPS and the best SVGP by 11% and 33%, and its divergence field is associated with subsequent drifter clustering, unlike SVGP's. All stages after the pre-registered comparison are disclosed revisions, reported with the failed attempts that motivated them.

1 Introduction

Submesoscale currents—fronts, filaments and vortices of 0.1–10 km that live for hours to days—carry much of the vertical exchange in the upper ocean and decide where buoyant material such as oil, plastics or larvae accumulates [McWilliams, 2016, D'Asaro et al., 2018]. They are too small and too fast for satellite altimetry; the most direct observations come from surface drifters released by the hundreds in dense arrays. The Lagrangian Submesoscale Experiment (LASER) deployed more than a thousand drifters in the northern Gulf of Mexico in winter 2016 [D'Asaro et al., 2018, Haza et al., 2018], and at hourly resolution its drogued drifters provide more than half a million velocity vectors. Turning this cloud of trajectory samples into a velocity map with honest uncertainty, and into the divergence field that controls clustering, is a spatio-temporal regression problem for which Gaussian processes (GPs) are the natural tool [Bretherton et al., 1976, Rasmussen and Williams, 2006].

Two obstacles stand in the way. The first is scale: a dense covariance of 10^6 scalar observations is out of reach, and the usual approximations—inducing points [Titsias, 2009, Hensman et al., 2013], nearest-neighbour or Vecchia factorisations [Vecchia, 1988, Datta et al., 2016, Katzfuss and Guinness, 2021]—trade exactness for speed. gp2Scale [Noack et al., 2023, 2025] showed that exact GPs scale to millions of points when the kernel is compactly supported, because the covariance matrix is then

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sparse and its Cholesky factor affordable. The second obstacle is structure. Velocity is a vector field whose components are tied together by kinematics: rotational eddies make the longitudinal velocity correlation exceed the transverse one, convergent fronts do the opposite, and wind-forced near-inertial motions rotate the velocity vector clockwise with a period of about a day. Every kernel of the gp2Scale family is scalar, non-negative and monotone within its support, so it must treat the two velocity components as unrelated scalars whose correlation can only decay.

This paper builds and tests the kernel this domain seems to demand: a compactly supported, positive semi-definite, matrix-valued kernel whose rotational and divergent parts are minus the Hessians of Wendland stream-function and velocity-potential covariances and whose third part is a rotary near-inertial covariance (Sec. 3). With a static sparsity pattern, gp2Scale-style distributed assembly, supernodal Cholesky factorisation and block Metropolis–Hastings, we fit it exactly to all 539,944 hourly velocity vectors (1,079,888 scalar observations) of LASER on 32 CPU cores, and compare it under an identical support and inference budget with the gp2Scale Wendland kernel and with SVGP on 48 drifters that were held out before any fitting.

The pre-registered comparison failed, and finding out why is the main result of this paper. With the pre-registered 5-km support, the kinematic kernel only tied the gp2Scale kernel on held-out drifters (RMSE 0.1459 vs $0.1456\,\mathrm{m\,s^{-1}}$), although it had won every pilot window. The training residuals show the reason: their dominant structured component is a clockwise near-inertial oscillation that stays coherent over at least 30 km, six times the support of the submesoscale kernels. Neither kernel could reach it, and three quarters of the squared error came from the fifth of held-out locations that no 5-km support reaches. Carrying the near-inertial band explicitly fixed this. Moving it into a local harmonic mean (a two-scale model) improved both kernels by 15% in RMSE without changing their ranking. Putting it back into the exact covariance—a rotary term on a 30-km, 12-h ellipsoid, whose Cholesky factor (45 GB) turned out to be affordable—gave the best model by a wide margin: RMSE 0.1186 and CRPS $0.0446\,\mathrm{m\,s^{-1}}$, 11% and 33% better than the best SVGP and 19% and 20% better than the pre-registered model. Its learned rotation frequency (chain median $1.03f$) sits at the peak of the drifters' rotary spectrum. Under the same support and protocol, a two-range version of the gp2Scale kernel, which can reach the same scales but cannot rotate, is 3.5% worse in RMSE and 7.3% worse in CRPS (drifter-bootstrap intervals excluding zero) and 178,153 nats lower in log likelihood: the covariance structure this domain demands is the rotation, and it pays only once its support reaches the scale on which the rotation is coherent. The exact posterior also resolves submesoscale kinematics with Rossby numbers of order one, and its divergence is associated with where drifters subsequently cluster ($\rho_s = -0.157$, against -0.114 for the comparator), a signal absent from the SVGP posterior.

The rest of the paper is organised as follows. Section 3 constructs the kernel, the two large-scale models and the exact inference pipeline; Section 4 describes the data, the protocol with its pre-registration and disclosed revisions, and the baselines; Section 5 reports the results of all stages and a reproduction of the gp2Scale MNIST benchmark; Section 6 analyses mechanism, failures, physical insight and limitations.

2 Related work

Scalable Gaussian processes. Inducing-point methods [Titsias, 2009, Hensman et al., 2013, Salimbeni et al., 2018] replace the covariance by a low-rank surrogate; structured interpolation [Wilson and Nickisch, 2015] exploits grids; nearest-neighbour and Vecchia methods [Vecchia, 1988, Datta et al., 2016, Katzfuss and Guinness, 2021, Wu et al., 2022] sparsify the precision, and SPDE models [Lindgren et al., 2011] do so through a discretised differential operator. Exact GPs have been scaled with conjugate gradients on GPUs [Wang et al., 2019, Gardner et al., 2018] and, closest to us, with compactly supported non-stationary kernels and sparse linear algebra in gp2Scale [Noack et al., 2023, 2025, Risser et al., 2024]. Covariance tapering [Furrer et al., 2006, Kaufman et al., 2008] multiplies a covariance by a compactly supported taper to approximate it, and the full-scale approximation [Sang and Huang, 2012] and multi-resolution approximations [Katzfuss, 2017] combine a coarse component with a sparse fine-scale component. Our two-scale model shares that architecture, but its coarse part is a physically structured local regression and its fine part an exact GP with a matrix-valued kernel; all of the scalable methods above work with scalar kernels.

Divergence- and curl-free kernels. Matrix-valued kernels obtained by differentiating scalar ones are classical in approximation theory [Narcowich and Ward, 1994, Fuselier, 2008] and appear in GP

models of magnetic fields [Wahlström et al., 2013] and of ocean currents: Berlinghieri et al. [2023] place squared-exponential priors on the stream function and velocity potential of drifter velocities and show improved current and divergence estimates on subsets of LASER and GLAD with hundreds to a few thousand observations. We make the construction compactly supported, add a rotary near-inertial term, run it exactly on the entire experiment, and find that at this scale the decisive structure is not the Helmholtz split but a rotary near-inertial covariance whose support reaches the scale on which near-inertial motion is coherent.

Ocean currents from Lagrangian data. Oceanographic objective analysis [Bretherton et al., 1976] and its locally stationary variants [Kuusela and Stein, 2018] interpolate observations with fitted covariances, typically on moving windows. Near-inertial oscillations are the most energetic high-frequency signal in drifter velocities [Pollard and Millard, 1970, Chaigneau et al., 2008, Alford et al., 2016] and are usually isolated by rotary spectra [Gonella, 1972] or demodulation [Bloomfield, 2000]; Lagrangian statistics such as along-trajectory autocovariances and relative dispersion are reviewed by LaCasce [2008]. Kinematic properties of LASER and GLAD flows have been estimated from drifter clusters [Berta et al., 2020], structure functions [Pearson et al., 2019] and convergence statistics [D'Asaro et al., 2018, Poje et al., 2014]. We use these diagnostics in a different role: to decide which part of the signal a compact covariance can carry at all.

3 Compactly supported kinematic and near-inertial kernels

3.1 Observation model

Each observation is a velocity vector $\mathbf{u}_i = (u_{x,i}, u_{y,i})$ measured by a drifter at position $\mathbf{s}_i \in \mathbb{R}^2$ (km, orthographic tangent plane centred at 27.5°N, 88.5°W) and time t_i (h); we write $z_i = (\mathbf{s}_i, t_i)$. We model

$$\mathbf{u}(z) = \mathbf{m}(z) + a(z)\,\mathbf{f}(z) + \boldsymbol{\varepsilon}, \qquad \mathbf{f} \sim \mathcal{GP}\big(\mathbf{0}, \mathbf{K}_\theta\big), \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}\big(\mathbf{0}, (\sigma_n^2 + e_i^2)\mathbf{I}_2\big), \tag{1}$$

where $\mathbf{K}_\theta(z, z') \in \mathbb{R}^{2\times 2}$ is a compactly supported matrix-valued kernel, e_i the velocity error reported with the data, $\mathbf{m}$ a large-scale component estimated from the other drifters (Sec. 3.4) and a a sampling-adaptive amplitude. Stacking both components of all $n = 539{,}944$ training locations gives a $2n \times 2n = 1{,}079{,}888 \times 1{,}079{,}888$ covariance matrix. The split of the signal into $\mathbf{m}$ and $\mathbf{f}$ is the central modelling decision of this paper: compact support makes the covariance of $\mathbf{f}$ sparse, but it also means that $\mathbf{f}$ can only carry structure that decorrelates within the support, so everything that is coherent over longer distances must be carried by $\mathbf{m}$.

3.2 Helmholtz decomposition with compactly supported potentials

In two dimensions a smooth velocity field decomposes into a rotational (divergence-free) and a divergent (curl-free) part, $\mathbf{u} = \nabla^\perp\psi + \nabla\phi$ with stream function ψ, velocity potential ϕ and $\nabla^\perp = (-\partial_y, \partial_x)$. Placing independent GP priors on ψ and ϕ induces a matrix-valued kernel for $\mathbf{u}$ through second derivatives of the potential covariances [Narcowich and Ward, 1994, Álvarez et al., 2012, Berlinghieri et al., 2023]. If the potential covariance is itself compactly supported, so is the velocity kernel. We use the Wendland function of the gp2Scale family [Wendland, 1995, Noack et al., 2025],

$$w(q) = (1-q)_+^8\,(32q^3 + 25q^2 + 8q + 1), \tag{2}$$

which is positive definite on $\mathbb{R}^3$ and C^6, so that second derivatives exist and are continuous. For a potential covariance $k(\Delta\mathbf{s}, \Delta t) = \frac{\sigma^2\rho^2}{22}\,w(q)$ with $q = \big(\|\Delta\mathbf{s}\|^2/\rho^2 + \Delta t^2/\tau^2\big)^{1/2}$, a direct calculation (Appendix A) gives the spatial Hessian $\nabla\nabla^\top k = \sigma^2\,[f_1(q)\,\mathbf{b}\mathbf{b}^\top - f_0(q)\,\mathbf{I}]$ with $\mathbf{b} = \Delta\mathbf{s}/\rho$ and

$$f_0(q) = -\frac{w'(q)}{22q} = (1-q)_+^7(16q^2+7q+1), \qquad f_1(q) = \frac{w''(q) - w'(q)/q}{22q^2} = 24(1+6q)(1-q)_+^6, \tag{3}$$

both finite at $q = 0$, where the normalisation makes σ^2 the per-component velocity variance. The velocity covariances of the two parts are

$$\mathbf{K}_\psi(z, z') = \sigma_\psi^2\Big[\big(f_0(q_\psi) - f_1(q_\psi)\|\mathbf{b}_\psi\|^2\big)\mathbf{I} + f_1(q_\psi)\,\mathbf{b}_\psi\mathbf{b}_\psi^\top\Big], \tag{4}$$

$$\mathbf{K}_\phi(z, z') = \sigma_\phi^2\Big[f_0(q_\phi)\,\mathbf{I} - f_1(q_\phi)\,\mathbf{b}_\phi\mathbf{b}_\phi^\top\Big], \tag{5}$$

each with its own spatial range ρ and temporal range τ. $\mathbf{K}_\psi$ is exactly divergence-free and $\mathbf{K}_\phi$ exactly curl-free. Their longitudinal and transverse correlations (velocity components parallel and perpendicular to the separation) are f_0 and $f_0 - f_1\|\mathbf{b}\|^2$ for $\mathbf{K}_\psi$ and swap for $\mathbf{K}_\phi$, so the kernel can express the excess of longitudinal over transverse correlation that marks rotational flow and the reverse that marks convergent fronts—something no pair of independent scalar kernels can do.

3.3 Rotary near-inertial term

Wind-forced near-inertial motions rotate the velocity vector clockwise (Northern Hemisphere) at a frequency close to the local Coriolis frequency f [Kunze, 1985, Alford et al., 2016]. If a, b are independent scalar GPs with covariance $\sigma_\iota^2 w(q_\iota)$ and $Q(\theta) = \left(\begin{smallmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{smallmatrix}\right)$, the process $Q(\omega t)(a, b)^\top$ has covariance

$$\mathbf{K}_\iota(z, z') = \sigma_\iota^2\, w(q_\iota)\, Q\big(\omega\,(t - t')\big), \tag{6}$$

whose auto-covariances oscillate as $\cos(\omega\Delta t)$ and whose u–v cross-covariance is antisymmetric, $\pm\sigma_\iota^2 w \sin(\omega\Delta t)$. The full kernel is

$$\mathbf{K}_\theta = \mathbf{K}_\psi + \mathbf{K}_\phi + \mathbf{K}_\iota, \qquad \theta = (\sigma_\psi, \rho_\psi, \tau_\psi, \sigma_\phi, \rho_\phi, \tau_\phi, \sigma_\iota, \rho_\iota, \tau_\iota, \omega), \tag{7}$$

positive semi-definite as a sum of covariances of explicit processes. Its support is the union of three space–time ellipsoids, so the covariance matrix is sparse; unlike every kernel of the gp2Scale family it is matrix-valued, sign-indefinite within its support and encodes the kinematics of the flow. Implementation: `code/oceangp/kernels.py` (`helmholtz_inertial`).

Reference kernel. The comparison model replaces Eq. (7) by the gp2Scale Wendland kernel with axis-aligned (ARD) ranges applied to each component independently, $\mathbf{K}^{\mathrm{W}} = \mathrm{diag}\big(\sigma_x^2 w(q_x), \sigma_y^2 w(q_y)\big)$ with $q_c^2 = (\Delta x/r_{x,c})^2 + (\Delta y/r_{y,c})^2 + (\Delta t/\tau_c)^2$ (Eq. 3 of Noack et al., 2025 with ARD ranges). Both kernels use the same amplitude $a(z)^2 = 1 + \gamma D_0/(D_0 + W(z))$, where $W(z)$ is the total weight of a Nadaraya–Watson smoother of the training data at z (local data support, $D_0 = 100$); this is the non-constant signal variance of Paciorek and Schervish [2006] and of the gp2Scale kernels and is positive semi-definite for any a (Theorem 1 of Noack et al., 2025).

3.4 The large-scale component

Pre-registered background. Following oceanographic objective analysis [Bretherton et al., 1976], the pre-registered model uses as $\mathbf{m}$ a Nadaraya–Watson (NW) smoother of the training velocities with Gaussian weights of scales $L = 25$ km and $T = 24$ h, chosen by leave-one-drifter-out error on training locations that the compact kernel cannot reach. To keep training and prediction consistent, $\mathbf{m}$ is *cross-fitted*: at a training location the contributions of the location's own drifter are removed, exactly as they are absent for an unseen drifter.

Near-inertial harmonic background (two-scale model). The analysis of Sec. 6 shows that the residual of this background is dominated by near-inertial motion that is coherent over tens of kilometres—a Gaussian window of $T = 24$ h attenuates the inertial frequency by $e^{-(fT)^2/2} \approx 5 \times 10^{-8}$ and therefore passes all of it to $\mathbf{f}$—and that a ratio estimator extrapolates single distant drifters where its data weight is tiny. The two-scale model replaces $\mathbf{m}$ by a local harmonic regression. Writing velocities as complex numbers $U = u_x + iu_y$, at every query point $z_q = (\mathbf{s}_q, t_q)$ we fit

$$U(\mathbf{s}, t) \;\approx\; \alpha + \beta\, e^{-if(t - t_q)} \tag{8}$$

by weighted least squares to the other drifters' observations with the same Gaussian weights $w_j(z_q)$, a ridge $\lambda \sum_j w_j |\beta|^2$ and a pseudo-weight κ that shrinks α towards the global mean velocity $\bar{U}$ and β towards zero:

$$\min_{\alpha,\beta\in\mathbb{C}} \sum_j w_j(z_q)\left|U_j - \alpha - \beta e^{-if(t_j - t_q)}\right|^2 + \lambda\Big(\sum_j w_j\Big)|\beta|^2 + \kappa\big(|\alpha - \bar{U}|^2 + |\beta|^2\big), \tag{9}$$

a 2×2 complex linear system per query point. The background is $\mathbf{m}(z_q) = \alpha + \beta$: α is the local sub-inertial flow and β the clockwise near-inertial velocity at time t_q, the kernel-weighted analogue of complex demodulation [Bloomfield, 2000] and of the slab picture of wind-forced inertial currents [Pollard and Millard, 1970]. Cross-fitting is as above, and f is the Coriolis frequency at the projection centre. With $\kappa = 0$ and $\beta \equiv 0$ the estimator reduces to the NW smoother. The bandwidths ($L = 25$ km,

$T = 24\,\mathrm{h}$, by the same isolated-location criterion), the shrinkage $\kappa = 1$ and $\lambda = 0.2$ were fixed by leave-one-drifter-out error on training drifters only (`code/oceangp/harmonic_background.py`). The amplitude covariate W is unchanged. Because $\mathbf{m}$ is a deterministic function of the training data that never sees a held-out drifter, the covariance of $\mathbf{f}$ and the exactness of the GP computation are unaffected.

3.5 Exact inference at scale

Static sparsity pattern and distributed assembly. gp2Scale distributes the covariance computation over block pairs and evaluates every entry, because the non-zeros are only known after evaluating the kernel. With a compact kernel and bounded prior ranges, however, all non-zeros lie inside the union of the *maximal* support ellipsoids. We therefore (i) re-index the locations along a space–time Morton curve and enumerate the candidate pairs once with a parallel cell list, and (ii) fix the upper-triangular CSC pattern of the $2n \times 2n$ matrix. For every hyperparameter proposal, Dask workers [Rocklin, 2015] (one in-process worker with 32 threads) evaluate 1,024 column strips of the kernel and write the values directly into the host buffer; entries outside the current support are stored as explicit zeros. Pruned entries are zero for every admissible θ, so nothing is approximated. In stages 1 and 2 both kernels share the maximal support $(5\,\mathrm{km}, 5\,\mathrm{km}, 24\,\mathrm{h})$, giving 1.62×10^8 location pairs and 1.29×10^9 stored entries (density 1.11×10^{-3}); in stage 3 the rotary term (and the comparator's long term) has its own $(30\,\mathrm{km}, 30\,\mathrm{km}, 12\,\mathrm{h})$ ellipsoid, and the union pattern has 4.3×10^8 location pairs and a Cholesky factor with 5.6×10^9 entries.

Exact log marginal likelihood. Because the pattern is static, the symbolic analysis of the supernodal Cholesky factorisation (CHOLMOD with METIS ordering; Chen et al., 2008, Karypis and Kumar, 1998) is computed once per process and reused; each likelihood evaluation performs one numeric factorisation $\mathbf{L}\mathbf{L}^\top = \mathbf{P}(\mathbf{A}\mathbf{K}\mathbf{A} + \mathbf{V})\mathbf{P}^\top$ (with $\mathbf{A} = \mathrm{diag}(a)$ and $\mathbf{V}$ the noise), an exact log-determinant $2\sum_j \log L_{jj}$ read from the supernode diagonals, and an exact solve. No stochastic trace or Lanczos estimate is involved.

Prediction and derived quantities. Posterior means use the sparse cross-covariance $\mathbf{K}_{X*}$; the 2×2 posterior covariance of a test location is $\mathbf{K}_{**} - \mathbf{W}^\top\mathbf{W}$ with $\mathbf{W} = \mathbf{L}^{-1}\mathbf{P}\mathbf{K}_{X*}$, i.e. one forward solve with a sparse right-hand side. Divergence and vorticity are linear functionals; a centred 50 m stencil applied to the exact posterior gives their posterior mean and variance at the cost of one additional forward solve per query point (Appendix B).

3.6 Hyperparameter inference

With 1.08×10^6 observations the posterior over θ is extremely concentrated, so a random-walk sampler started from small-window pilot values would spend its whole budget in burn-in. We therefore (i) locate the posterior mode with Nelder–Mead in log-coordinates at full scale, (ii) obtain proposal scales from a diagonal finite-difference curvature, and (iii) run the block Metropolis–Hastings sampler of gp2Scale (fvGP `gpMCMC` with adaptive `ProposalDistribution` blocks; Haario et al., 2001, Shaby and Wells, 2010) with blocks $\{\sigma_\psi, \rho_\psi, \tau_\psi\}$, $\{\sigma_\phi, \rho_\phi, \tau_\phi\}$, $\{\sigma_\iota, \rho_\iota, \tau_\iota, \omega\}$, $\{\sigma_n, \gamma\}$ (reference kernel: $\{x\}$, $\{y\}$, $\{\sigma_n, \gamma\}$), uniform priors on log-parameters inside physically motivated bounds, and 100 iterations. Predictions use the chain sample with the highest posterior density. The run is checkpointed so that it survives the two-hour limit of our job scheduler: every evaluated (hyperparameter, log-likelihood) pair is cached, the Nelder–Mead path is replayed deterministically, and the sampler state is pickled after every iteration.

4 Experimental setup

Data. We use the quality-controlled LASER release of 15-min interpolated trajectories [D'Asaro et al., 2017] (3,564,195 records of 1,432 trajectory segments; drogue-loss splitting by Haza et al., 2018; low-pass filtering by Yaremchuk and Coelho, 2015). We keep drogued segments, records on the full hour (15-min records are filtered near-duplicates), velocity errors $\leq 0.1\,\mathrm{m\,s^{-1}}$ and positions inside the Gulf of Mexico, project positions onto the tangent plane and rotate velocities into its fixed frame. This yields 568,295 velocity vectors from 956 drifters between 20 January and 30 April 2016 (Fig. 1). The source DOI, file checksum and every filtering count are logged (`logs/data_prep/`).

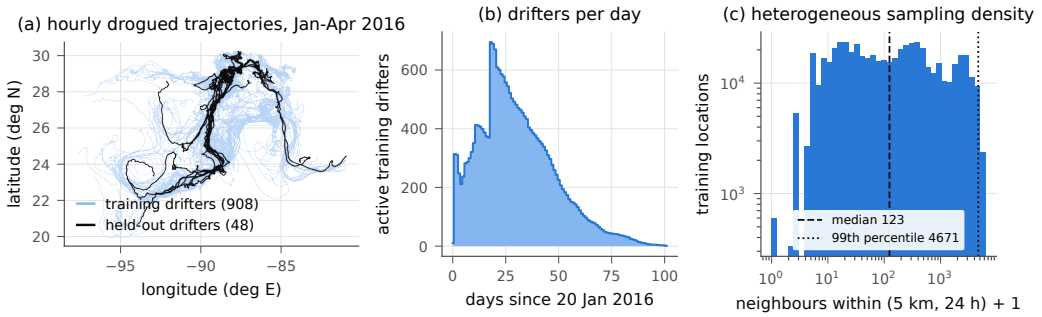


Figure 1: The LASER data set. (a) Hourly drogued trajectories of the 908 training drifters and the 48 held-out drifters. (b) Number of active training drifters per day; the dense deployments of late January and early February are followed by dispersal. (c) Number of other training locations within the $(5\,\mathrm{km}, 24\,\mathrm{h})$ support of each training location (median 123, 99th percentile 4,671): the row densities of the covariance span four orders of magnitude.

Held-out drifters. Before any model was fitted, 48 of the 956 physical drifters (5%) were drawn with seed 20160121 and removed entirely; their 28,351 velocity vectors (56,702 scalar observations, 5.25% of the training size) form the test set. Holding out whole drifters measures what matters for mapping—prediction at an independent instrument—and avoids the trivial interpolation between consecutive hourly records of one trajectory. All pilot studies, bandwidth choices and baseline tuning used validation drifters drawn from the 908 training drifters only.

Protocol. The domain, task, kernel, hypothesis (the gp2Scale kernel attains lower marginal likelihood and worse held-out RMSE, CRPS and calibration under an identical support and inference budget), split, metrics, support budget and inference protocol were committed to a version-controlled pre-registration before any full-scale run (`method_plan.md`). Four disclosed amendments to the comparison followed (Addenda B–E; Addendum A only made the runs resumable and recorded a late pilot result); we report all of their results. *Stage 1* is the pre-registered comparison under the NW background. A *budget-matched continuation* (+60 BMH iterations per chain, decided during the stage-1 chains and before any test prediction) removes an asymmetry of our own making: the pre-registered bounds of the rotational and divergent terms ((5 km, 12 h) and (3 km, 8 h)) were tighter than the static pattern, whereas each gp2Scale-kernel component could use all of it. *Stage 2* replaces the NW background by the near-inertial harmonic background for both kernels (Addendum C); *stage 2b* repeats the adapted kernel with the lower bound of ω removed, selected against stage 2 by training likelihood (Addendum D); *stage 3* returns to the NW background with shrinkage and gives the rotary term its own $(30\,\mathrm{km}, 30\,\mathrm{km}, 12\,\mathrm{h})$ ellipsoid, after measuring that such supports are affordable (Addendum E). Stages 2–3 were designed after the stage-1 test errors had been examined, from training-only diagnostics, and each was fixed and committed before any of its own runs; no stage-2 test result was read before stage 3 was committed.

Models compared. (i) *Helmholtz–inertial* (ours): Eq. (7) with mean, amplitude and noise model, 12 hyperparameters. (ii) *Wendland-ARD*: the gp2Scale kernel with ARD ranges per component, the same mean, amplitude, noise model, sparsity pattern, optimisation budget (Nelder–Mead 150 vs 120 evaluations, as pre-registered for the larger parameter count) and sampler, 10 hyperparameters. (iii) In stage 3, the comparator is a *two-range Wendland-ARD* kernel—the sum of two gp2Scale Wendland-ARD kernels per component, one on each ellipsoid of the same union support—with 18 hyperparameters, Nelder–Mead budget 160 and five BMH blocks. (iv) *SVGP* [Hensman et al., 2013] in GPyTorch [Gardner et al., 2018]: two independent latent GPs with ARD Matérn-3/2 kernels on standardised (x, y, t), natural-gradient variational updates [Salimbeni et al., 2018], Adam for hyperparameters, mini-batches of 4,096 locations, trained on the full training set. Kernel, number of inducing points and the regression target (raw velocities, or the residual of either background) were selected on the validation split (Table 6): 2,000 inducing points per component beat 1,000, and modelling the residual of the near-inertial harmonic background beat raw velocities, which in turn beat the NW residual. The best validated configuration (harmonic residual) is our baseline row; for

transparency we also report SVGP on raw velocities and on the NW residual, each with three seeds. (v) The background fields alone.

Metrics. On all 56,702 scalar test observations we report RMSE, the continuous ranked probability score (CRPS) of the Gaussian predictive distribution of the observation [Gneiting and Raftery, 2007], the empirical coverage of central credible intervals at levels $0.1, \ldots, 0.9, 0.95, 0.99$ (the coverage curve), and the negative log predictive density, together with breakdowns by local data support. Uncertainty of paired differences is assessed with a drifter-level bootstrap (2,000 resamples of the 48 held-out drifters).

Hardware and accounting. Every run was pinned to 32 cores of one socket of an Intel Xeon node and used at most 256 GB of memory; independent runs used disjoint core sets. Linear algebra used MKL and CHOLMOD; the Dask cluster ran 32 worker threads. Wall-clock times include Nelder–Mead, curvature, sampling and prediction; peak memory is the maximum resident set size over all stages.

Pipeline validation. Before trusting the drifter results we reproduced the MNIST benchmark of Noack et al. [2025] with the provided fvGP code (Sec. 5.1).

5 Experimental results

Exact inference on the full training set. Every likelihood evaluation of every stage factorised the full $1{,}079{,}888 \times 1{,}079{,}888$ covariance. The $(5\,\mathrm{km}, 24\,\mathrm{h})$ static pattern of stages 1 and 2 holds 1.29×10^9 entries (density 1.11×10^{-3}), of which 82% are non-zero at the adapted kernel's MAP and 37% at the gp2Scale kernel's (whose u–v blocks vanish identically); its supernodal Cholesky factor has 1.77×10^9 entries, a numeric factorisation takes 43 s on 32 cores (median over all evaluations), and training peaks at 41.6 GB. The stage-3 pattern, the union with a $(30\,\mathrm{km}, 12\,\mathrm{h})$ ellipsoid, holds 3.44×10^9 entries (density 2.95×10^{-3}) and its factor 5.63×10^9; a factorisation takes 139 s, training peaks at 116.6 GB, and prediction with full 2×2 predictive covariances at 197.4 GB. Table 1 summarises the held-out results of all stages, Fig. 2 their calibration, Fig. 4 the learned kernels and the sparsity, and Fig. 5 the chains.

Pre-registered comparison (stage 1). The hypothesis was not supported. With the pre-registered background, the adapted kernel reached a lower MAP log marginal likelihood than the gp2Scale kernel (by 20,679 nats) and a statistically indistinguishable but slightly worse held-out score: RMSE 0.1459 vs 0.1456 $\mathrm{m\,s^{-1}}$ and CRPS 0.0559 vs 0.0556 $\mathrm{m\,s^{-1}}$, with drifter-bootstrap differences of +0.0003 $[+0.0001, +0.0005]$ and +0.0003 $[+0.0000, +0.0005]$. Removing the bound asymmetry and adding 60 iterations per chain (the budget-matched continuation; Table 3) did not change this. This contradicts the pilot study, in which the adapted kernel won all 6 window-and-mean combinations in marginal likelihood (by 4,697 to 30,086 nats) and in validation CRPS, and 5 of them in validation RMSE (Appendix D).

Two-scale model (stages 2 and 2b). Replacing the background by the near-inertial harmonic estimate (Eq. 9) improved both kernels by about 15% in RMSE and 11% in CRPS (gp2Scale kernel: 0.1456→0.1234 and 0.0556→0.0494; drifter-bootstrap RMSE difference -0.0222 $[-0.0472, -0.0068]$), with the largest gains where no training drifter is nearby (0.280→0.233 $\mathrm{m\,s^{-1}}$ at isolated locations). The kernel ranking did not change. The adapted kernel's rotary frequency moved to its pre-registered lower bound; with the bound removed (stage 2b, selected by training likelihood) it settled at $\omega = 0.62f$ (median over the second half of the chain) and the adapted kernel reached RMSE 0.1238 and CRPS 0.0496, against 0.1234 and 0.0494 for the gp2Scale kernel (differences +0.0004 $[+0.0003, +0.0006]$ and +0.0003 $[+0.0001, +0.0004]$).

Near-inertial covariance (stage 3). Giving the rotary term a 30-km, 12-h ellipsoid—and putting the near-inertial band back into the covariance instead of the mean—produced the best model of all stages by a wide margin in every score: log marginal likelihood 1,928,053 (324,395 nats above the best two-scale model), held-out RMSE 0.1186 and CRPS 0.0446, 90% coverage 0.930, and NLPD -1.443. Against the best two-scale model the drifter-bootstrap differences are -0.0048 $[-0.0086, -0.0023]$ in RMSE and -0.0048 $[-0.0063, -0.0034]$ in CRPS; the gain is largest at locations with 10–99 training

Table 1: Held-out results on all 56,702 scalar velocity observations of the 48 held-out drifters. $|\mathcal{D}|$ counts scalar observations (two per location); every exact-GP row factorises the full $|\mathcal{D}_{\text{train}}| \times |\mathcal{D}_{\text{train}}|$ covariance. Coverage: empirical coverage of central 90% predictive intervals (full curves in Fig. 2). Wall-clock: Nelder–Mead, curvature, 100 BMH iterations and test prediction on 32 cores; peak memory: maximum resident set size over all stages. SVGP rows: mean $\pm$ standard deviation over three seeds; the harmonic-residual row is the validated baseline. Stage 1 is the pre-registered comparison, stages 2 and 3 are the disclosed revisions of Sec. 4. The stage-2 adapted row is the free-ω variant selected by training likelihood (Addendum D); its wall-clock is that of the identical-protocol bounded run, whose evaluations it shared.

Method	Kernel	$\lvert\mathcal{D}_{\text{train}}\rvert$	$\lvert\mathcal{D}_{\text{test}}\rvert$	RMSE	CRPS	Coverage	Wall-clock (h)	Peak memory (GB)
Exact GP, stage 3	Helmholtz–inertial (ours) with 30-km rotary term	1,079,888	56,702	0.1186	0.0446	0.930	20.1	197
Exact GP, stage 3	two-range Wendland-ARD (gp2Scale family)	1,079,888	56,702	0.1229	0.0481	0.938	26.1	167
Exact GP, stage 2	Helmholtz–inertial (ours)	1,079,888	56,702	0.1238	0.0496	0.944	7.8	71
Exact GP, stage 2	Wendland-ARD (gp2Scale)	1,079,888	56,702	0.1234	0.0494	0.945	6.8	64
Exact GP, stage 1 (pre-reg.)	Helmholtz–inertial (ours)	1,079,888	56,702	0.1459	0.0559	0.944	7.3	68
Exact GP, stage 1 (pre-reg.)	Wendland-ARD (gp2Scale)	1,079,888	56,702	0.1456	0.0556	0.944	6.8	60
SVGP, harmonic residual	Matérn-3/2 ARD, M=2000	1,079,888	56,702	0.1336±.0016	0.0667±.0006	0.903±.003	0.7	4
SVGP, velocities	Matérn-3/2 ARD, M=2000	1,079,888	56,702	0.1398±.0005	0.0725±.0003	0.908±.001	0.5	4
SVGP, NW residual	Matérn-3/2 ARD, M=2000	1,079,888	56,702	0.1579±.0008	0.0765±.0002	0.909±.002	0.6	4

observations in the 5-km support (RMSE 0.0951→0.0694), and at isolated locations the harmonic mean remains slightly better (+0.0039 $[+0.0006, +0.0069]$). The rotary term used its whole support (range 28.5 km, time scale at the 12-h cap) and its frequency settled at $\omega = 1.03f$ (median over the second half of the chain; range 0.90–$1.10f$, $0.90f$ at the MAP), the clockwise peak of the drifters' rotary spectra ($1.025f$). The decisive comparison is with the paper's kernel family under the same support, mean and protocol. The two-range Wendland-ARD comparator, whose long term may use the same 30-km, 12-h ellipsoid but cannot rotate, ended 178,153 nats lower in log likelihood—a gap that its chain did not close although it climbed beyond hand-set probes at the adapted kernel's ranges (1,674,486 nats; Appendix E)—and reached RMSE 0.1229 and CRPS 0.0481. The adapted kernel is better by -0.0043 $[-0.0064, -0.0029]$ in RMSE and -0.0035 $[-0.0044, -0.0026]$ in CRPS, and in every data-support regime except the densest, where the two tie (Fig. 3). Long support without rotation helped the gp2Scale kernel little over its two-scale version (RMSE 0.1234→0.1229).

Against SVGP. The best SVGP configuration—Matérn-3/2 with 2,000 inducing points per component on the residual of the same harmonic background, selected on validation drifters—reaches RMSE 0.1336 ± 0.0016 and CRPS 0.0667 ± 0.0006 over three seeds, and a larger inducing set does not help (Appendix H). The stage-3 adapted kernel beats it by -0.0168 $[-0.0258, -0.0101]$ in RMSE and -0.0228 $[-0.0263, -0.0197]$ in CRPS, with an NLPD of -1.443 against -0.643 per scalar observation, and so do all two-scale exact GPs (gp2Scale kernel: -0.0119 $[-0.0199, -0.0063]$ and -0.0180 $[-0.0218, -0.0145]$). The advantage is concentrated where data are dense: with ≥ 1000 training locations in the support, the exact GPs' RMSE is 0.0185 against 0.0715 for SVGP, while at isolated locations the two are statistically indistinguishable (0.237 vs 0.231; difference +0.0065 $[-0.0045, +0.0158]$; Fig. 3). With the pre-registered background the exact GPs had lost to SVGP

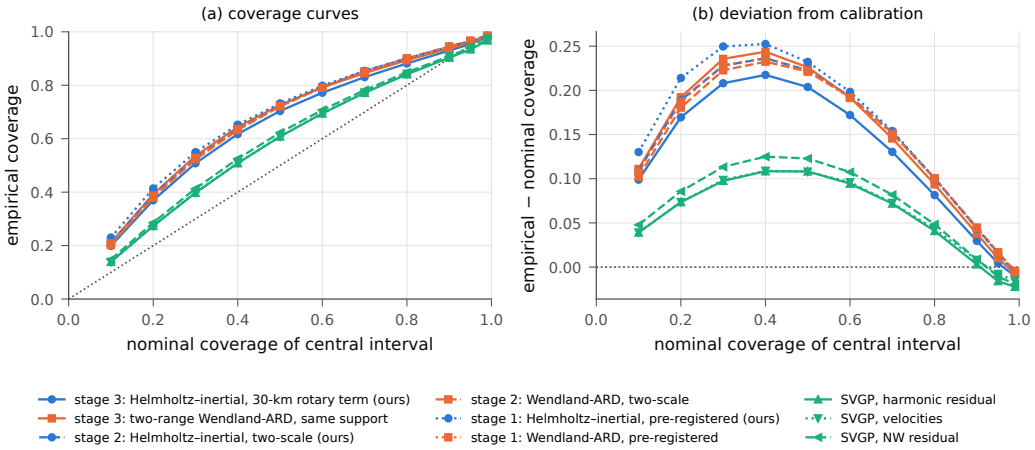


Figure 2: Calibration on the held-out drifters for every method of Table 1: (a) empirical coverage of central predictive intervals against their nominal level, (b) the deviation from it (SVGP: mean over three seeds, whose spread is at most 0.004). All exact GPs are over-dispersed up to the 95% level and slightly under-dispersed at 99%, the signature of a mixture of near-perfect interpolation inside clusters and heavy-tailed extrapolation errors (Sec. 6.2); the stage-3 adapted kernel is the closest of them to the diagonal up to 95%. SVGP is closer at low levels but under-covers at 95% and 99%.

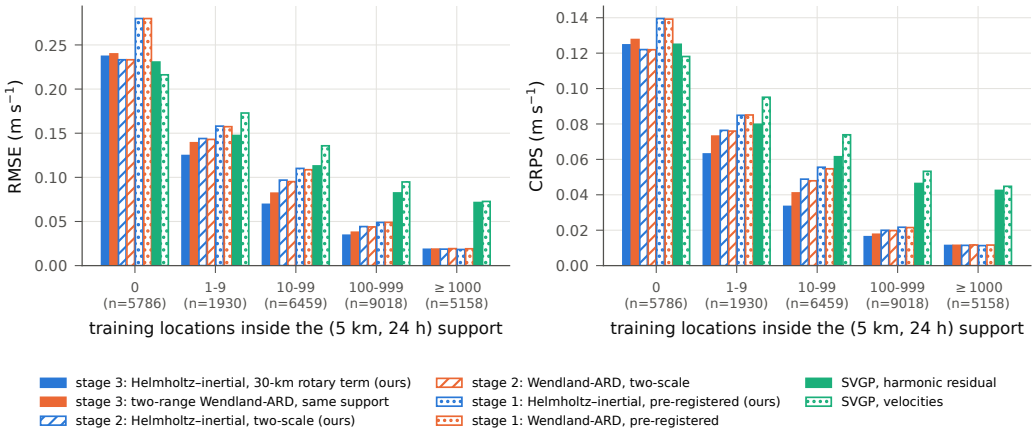


Figure 3: Held-out RMSE and CRPS by local data support, i.e. the number of training locations inside the $(5\,\mathrm{km}, 24\,\mathrm{h})$ support of the test location (filled: stage-3 models and the validated SVGP; hatched: stage-2 and stage-1 models and SVGP on raw velocities). The exact GPs are up to four times more accurate than SVGP inside dense clusters; the stage-3 adapted kernel gains most at intermediate support (1–99 locations), and at isolated locations everything rests on the large-scale component.

at isolated locations by a wide margin (0.280 vs 0.216), which had cost them the overall RMSE comparison.

Calibration. With their 90% intervals the stage-3 models cover 0.930 (adapted kernel) and 0.938 (comparator) of the held-out observations, the two-scale models 0.944 and 0.945, and SVGP 0.903; at the 95% level the exact GPs cover between 0.954 and 0.967 and SVGP 0.934 (Fig. 2; every curve, including the secondary runs, is tabulated in Appendix F). No model is calibrated across levels. The exact GPs' intervals are too wide inside dense clusters and too narrow at isolated locations (Sec. 6.2), which the aggregate curves show as over-dispersion up to the 95% level (at the 10% level the stage-3 adapted kernel covers 0.199 of the observations) and under-dispersion at 99% (0.980). Of the exact GPs, the stage-3 adapted kernel is closest to the diagonal at every level up to 95% and furthest from it at 99%.

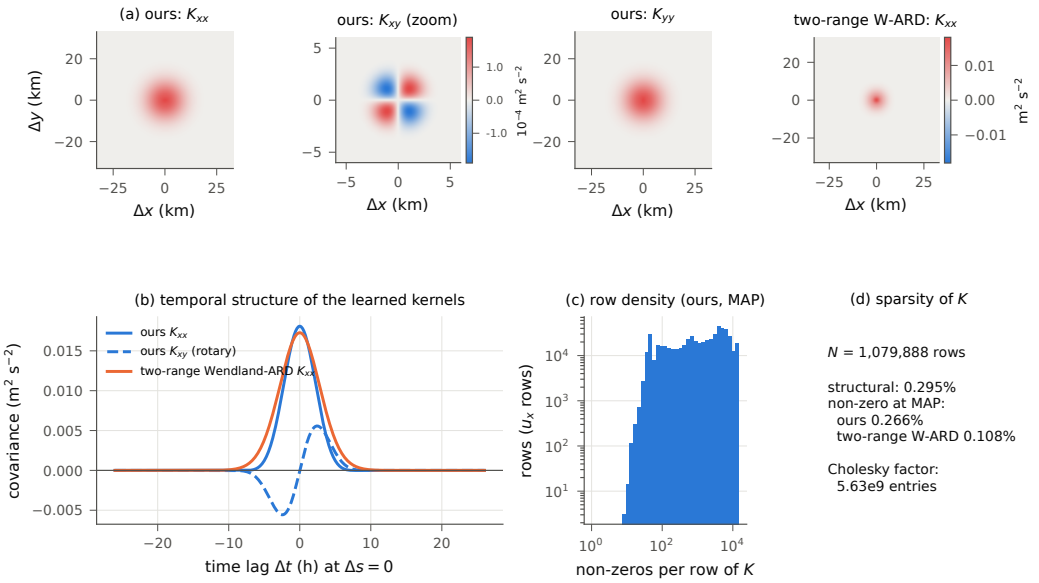


Figure 4: Learned stage-3 kernels and covariance sparsity. (a) 2×2 blocks of the adapted kernel at zero time lag, with K_{xy} zoomed to the 5-km support of the Helmholtz terms, where it shows the quadrupole of rotational flow, and K_{xx} of the two-range Wendland-ARD comparator on the same colour scale as the adapted kernel's diagonal blocks. (b) Temporal structure at zero separation: the adapted kernel's rotary term produces an antisymmetric u–v covariance; the comparator can only decay. (c) Distribution of non-zeros per row of the adapted kernel's covariance at the MAP. (d) Structural density of the stage-3 pattern, density at the MAP for both kernels, and size of the Cholesky factor.

Chains, cost and sparsity. Fig. 5 shows the optimisation and sampling traces of the stage-3 runs. Nelder–Mead covered most of the climb; the BMH chains then raised the log likelihood by a further 39,377 (adapted kernel) and 95,214 nats (comparator), with cumulative per-block acceptance rates between 0.07 and 0.40 and between 0.24 and 0.35, and both were still climbing after 100 iterations (Sec. 6.2). The comparator's larger gain reflects its lower starting point: its chain ended 178,153 nats below the adapted kernel's. A complete stage-3 run—Nelder–Mead, curvature, sampling and prediction of 28,351 held-out locations with full 2×2 predictive covariances—took 20.13 (adapted kernel) and 26.12 h (comparator) of wall-clock on 32 cores, of which prediction took 2.38 h; the two-scale runs took 6.76–7.85 h. At the MAP, the adapted kernel's covariance is non-zero on 90% of the stage-3 pattern, i.e. on a fraction 2.66×10^{-3} of all $1{,}079{,}888^2$ entries, the comparator's on 37% because its u–v blocks vanish (Fig. 4); the rows of the adapted kernel's covariance hold between 9 and 15,289 non-zeros (median 1,418), the dense rows being those of drifter clusters.

5.1 Validating the exact-GP pipeline on the gp2Scale MNIST benchmark

To check that our use of the gp2Scale machinery is sound before trusting the drifter results, we reproduced the MNIST experiment of Noack et al. [2025] (regression of $\Pr(y = 5)$ on 784 pixels, random 60,000/10,000 splits, Brier score) with the provided fvGP code: `GP(gp2Scale=True)` with its Dask block assembly, the kernel $k_{\mathcal{W}} k_d$ (Wendland on the ℓ_1 distance times the nearest-neighbour delta kernel), fvGP's `gpMCMC` sampler (120 iterations) and three seeds (Table 8). The paper does not specify the delta kernel beyond its role, and two literal readings failed (a fixed-radius ℓ_1 Wendland is uninformative at sparse radii and not positive definite at small noise; `attempts_log.md`, A8); we used normalised counts of shared k-nearest-neighbour balls with k chosen by marginal likelihood on the first split only. The resulting covariance has 0.32% non-zeros. Our Brier score is 0.0102 ± 0.0003 (accuracy 98.99%), *lower* than the reported 0.018 ± 0.002 and therefore outside the pre-registered tolerance $[0.014, 0.022]$. As a check of the data handling and scoring, we also trained SVGP with the paper's configuration (500 inducing points, RBF kernel) on the same splits: it reaches a Brier score of 0.0135 ± 0.0006, likewise far below the reported 0.033, while the ordering of the two methods (exact

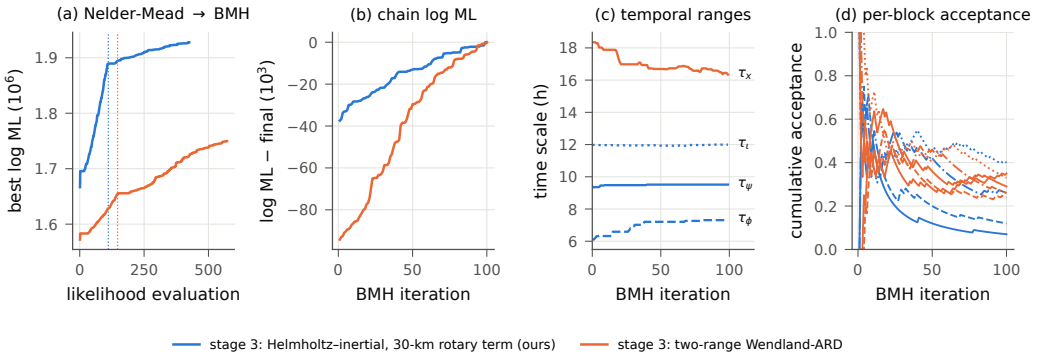


Figure 5: Hyperparameter inference at full scale (stage 3). (a) Best log marginal likelihood against the number of exact likelihood evaluations (dotted lines: end of Nelder–Mead). (b) Log likelihood of the BMH chain relative to its final value. (c) Temporal ranges of the adapted kernel's three terms (the rotary τ_ι at its 12-h cap) and of the comparator's short-range u_x term (its long-range terms sit at the same cap). (d) Cumulative per-block acceptance rates (one line per block).

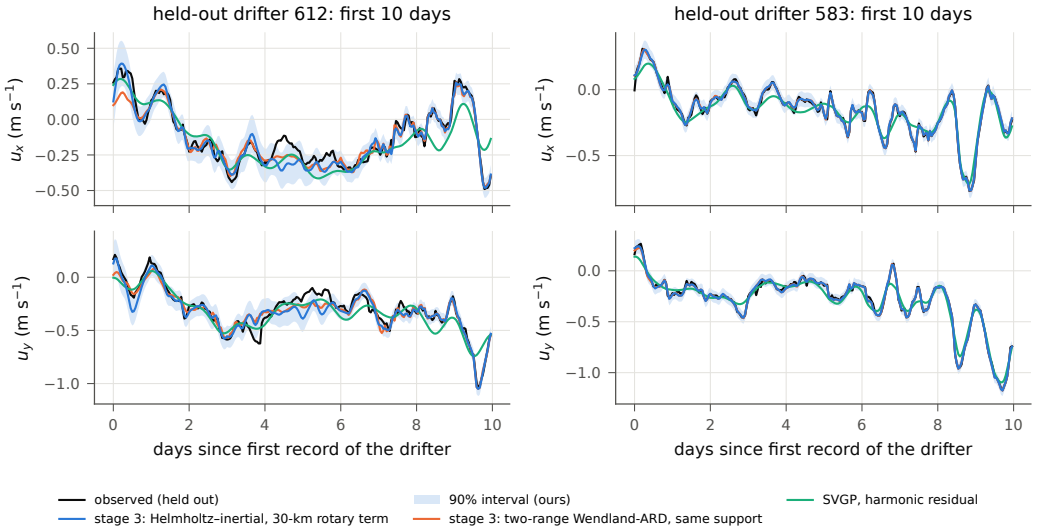


Figure 6: Predictions for the two held-out drifters with the most records (first ten days): observed velocity components, posterior means of the stage-3 exact GPs and of SVGP, and the adapted kernel's 90% predictive interval. Where the drifter is surrounded by training drifters the exact GPs follow hour-scale fluctuations that SVGP smooths out; where it is alone the interval widens.

gp2Scale better than SVGP) is preserved. The gap to the reported numbers is therefore common to both methods and points to a difference in the experimental setup that the paper does not let us resolve, rather than to a defect specific to our gp2Scale build. We read this as a faithful pipeline whose MNIST score depends on an under-specified component of the kernel, not as evidence against the reproduction, but we report it as a miss of the stated criterion. Independently, on the drifter data our log marginal likelihood agrees with fvGP's own gp2Scale evaluation to a relative error of 6.2×10^{-15}.

6 Analysis

The results raise three questions: which structure of the drifter velocities a compactly supported covariance can carry at all (Sec. 6.1), where and why the exact GPs fail (Sec. 6.2), and what the exact posterior reveals about the flow that an approximate one does not (Sec. 6.3). Every statement below

rests on a logged artefact: fitted hyperparameters, covariance diagnostics of the training residuals, regime and bootstrap breakdowns of the held-out errors, and derived posterior fields.

6.1 Mechanism: what a compact covariance can and cannot carry

Why a kernel that won every pilot tied at full scale. Three logged quantities explain the stage-1 outcome. *(i) The variance budget.* At the MAP, the inertial term carries 77% of the adapted kernel's prior variance (rotational 22%, divergent 1%) and 97% of the variance of its posterior-mean anomaly at held-out locations; within the 5 km support this term is an isotropic, rotating Wendland kernel, so most of the adapted kernel's spatial structure is that of a scalar Wendland kernel (Fig. 4). *(ii) What the inertial term is asked to fit.* The residual of the pre-registered background is dominated by near-inertial motion coherent far beyond the support (Fig. 7a): the equal-time cross-drifter correlation is still 0.31 at 20–30 km, and the antisymmetric part at a 6-hour lag lies between -0.34 and -0.30 for every separation band from 5 to 30 km—a clockwise rotation at the inertial period (rotary spectra peak at $1.025f$ with clockwise/anticlockwise power ratio 5.3). A compact kernel can only represent the part of this coherence that lies within its support, so both kernels push every range to the cap and still decorrelate within a few kilometres (Fig. 8c). Along trajectories, where the residual correlation at a 3-hour lag is 0.80, the two fitted kernels imply 0.08 and 0.10 (Fig. 8a), because a drifter moves 3–5 km in three hours. The adapted kernel does reproduce the sign and short-lag size of the rotary cross-covariance, which the gp2Scale kernel sets to zero by construction (Fig. 8b), but only for lags of a few hours. *(iii) Where a kernel can matter at all.* Three quarters of the squared error arise where no kernel reaches. Among covered locations the adapted kernel is significantly better in CRPS where data are densest (≥ 1000 training locations in the support: $\Delta\mathrm{CRPS} = -0.0003$, 95% drifter-bootstrap interval $[-0.0005, -0.0000]$) and significantly worse at 10–99 locations ($\Delta\mathrm{RMSE} = +0.0015$, $[+0.0008, +0.0022]$), where a near-inertial term truncated at 5 km mis-weights distant and lagged neighbours.

Where the near-inertial band lives decides what the kernel can do. The rotary term's learned frequency is a direct readout of what it is fitting. Truncated at 5 km with the near-inertial band in the residual (stage 1), it settled at $\omega = 0.85f$; with the band moved into the mean (stage 2b) it drifted to $0.62f$, a slow rotation with no inertial content left to model; given a 30-km support (stage 3) it settled at $1.03f$ (MAP $0.90f$), at the drifters' rotary-spectrum peak. Only in stage 3 does the adapted kernel represent the structure it was designed for: along trajectories it implies a correlation of 0.32 at a 3-hour lag (stage 1: 0.08; data: 0.80) and an antisymmetric u–v covariance of the data's magnitude, though compressed in time by the 12-h support (minimum -0.27 at a 2-hour lag against the data's -0.27 at 6 hours; Fig. 8). The two-scale mean and the near-inertial covariance are two answers to the same diagnosis—the compact covariance must not be asked to carry coherent motion it cannot reach—and the covariance answer is the better one, because it treats the near-inertial field as a random process with its own uncertainty and lets each held-out location borrow near-inertial information, with the correct phase, from drifters up to 30 km and 12 h away. This is exactly where the gain appears: at locations with 1–99 training observations in the 5-km support, i.e. where the submesoscale kernel alone is starved but other drifters are within near-inertial range. The stage-3 comparator isolates the role of rotation. With the same 30-km, 12-h ellipsoid but no rotation, the two-range Wendland-ARD kernel gains along-trajectory memory (0.24 at a 3-hour lag) but no rotary covariance, ends 178,153 nats below the adapted kernel, and improves held-out RMSE only from 0.1234 to 0.1229 over its two-scale version. Range is necessary; rotation is what the near-inertial covariance needs.

Where kinematic structure matters. Across stages 1 and 2 the same regime pattern recurs: the adapted kernel is significantly better in CRPS in the densest regime (stage 2: -0.0002, $[-0.0003, -0.0001]$) and significantly worse at 10–99 locations ($+0.0017$ in RMSE, $[+0.0012, +0.0022]$). At full scale the data assign the divergence- and curl-free terms only a few per cent of the prior variance (stage 3: rotational 4%, divergent 1%), whereas the 72-hour pilot fits found a short-lived (about 1 h), kilometre-scale divergent component in both windows. We read this as non-stationarity rather than as absence of kinematic structure: a single set of hyperparameters for 100 days and the whole northern Gulf averages frontal-scale divergence that dense windows resolve into a broad, weak term, and the derived Helmholtz kernels—whose correlations change sign within the support—are penalised wherever the data do not resolve that sign change.

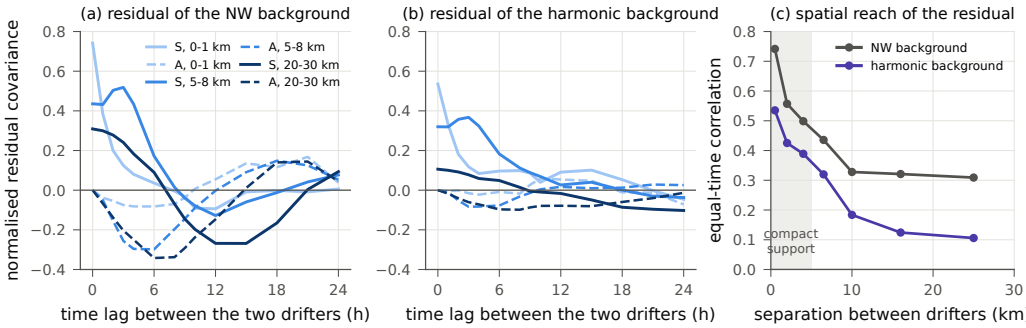


Figure 7: Near-inertial coherence of the training residuals (no model involved). Symmetric (S, solid) and antisymmetric (A, dashed; negative = clockwise rotation) parts of the cross-drifter residual covariance against time lag for three separation bands, (a) for the residual of the pre-registered NW background and (b) for that of the near-inertial harmonic background; (c) equal-time correlation against separation (grey: 5-km support of the submesoscale kernels).

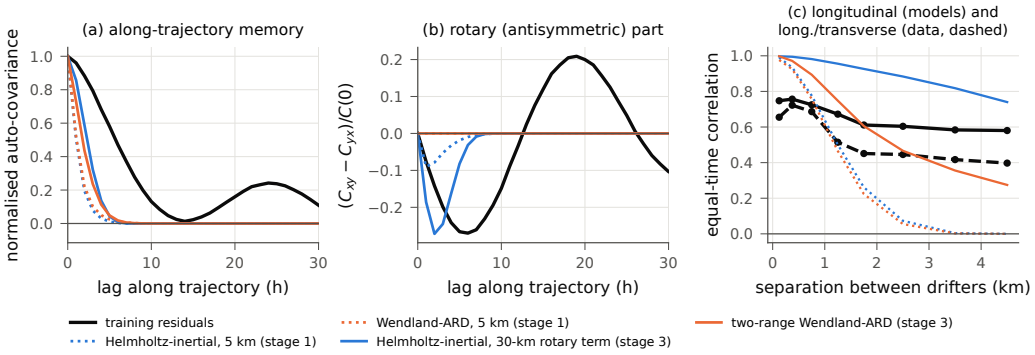


Figure 8: Second-order structure of the training residuals against the structure implied by the fitted kernels. (a) Along-trajectory auto-covariance; (b) its antisymmetric (rotary) part; (c) equal-time longitudinal (solid) and transverse (dashed) correlation between different drifters. Stage-1 kernels (dotted; 5-km support) decorrelate within a few hours along trajectories, and the stage-1 adapted kernel reproduces the rotary signature only at lags of a few hours; the stage-3 adapted kernel (solid blue; 30-km, 12-h rotary term) carries much more of both. The two-range Wendland-ARD comparator (solid orange) gains memory but, by construction, no rotary part.

6.2 Failure analysis

The data frontier. 20.4% of the held-out locations have no training observation inside the $(5\,\mathrm{km}, 24\,\mathrm{h})$ support. Under the 5-km kernels the posterior mean there is the background and the posterior variance the amplitude-inflated prior, and these locations carry 75% of the squared error of the pre-registered adapted model (75% for the gp2Scale kernel; Fig. 3). The stage-3 near-inertial covariance reaches some of them, but because its gains are concentrated at intermediate support, their share of its squared error rises to 82%. Five held-out drifters account for 69% of it, and one alone—a drifter that left the experiment's core and entered the sparsely sampled Loop-Current region—for 28.5%: its RMSE is $0.42\,\mathrm{m\,s^{-1}}$ against 0.20 for SVGP. The Nadaraya–Watson background is a ratio estimator; at a median data weight of 1.1 where this drifter's errors exceed $0.6\,\mathrm{m\,s^{-1}}$ it extrapolated the velocity of a single distant fast drifter, whereas SVGP's posterior reverts to its prior mean. The same pathology is visible in training-only leave-one-drifter-out errors at low data weight (RMSE $0.331\,\mathrm{m\,s^{-1}}$), which is how the stage-2 shrinkage was selected. With the shrinkage of the two-scale background, the same drifter's RMSE falls to $0.25\,\mathrm{m\,s^{-1}}$ and its share of the squared error to 13.8%.

Supports and the cost of exactness. Every spatial range of the submesoscale terms ended at the 5 km cap of the static pattern in every stage, while the residual stays correlated at 0.31 at 20–30 km

(Fig. 7). Table 7 measures what raising the cap costs. Location pairs grow only $4.6\times$ from 5 to 30 km, because most data sit in a few clusters that are fully connected already, and the Cholesky factor grows in proportion to the pattern (1.36–1.39 entries per stored entry). The union of the 5 km/24 h ellipsoid with a 30 km/12 h near-inertial ellipsoid needs a 5.6×10^9-entry factor (45 GB) and, in the stage-3 runs, 139 s per factorisation. Exactness itself is therefore affordable at near-inertial scales on one node; what grows is the cost of the hyperparameter search, which needs several hundred factorisations per run (425 and 572 in stage 3, i.e. 17.75 and 23.74 h of training). We fixed the 5 km budget for the time per likelihood evaluation and only measured these costs after stage 2, which is why the near-inertial covariance became a separate stage. Before we forced 64-bit indices, supports of $(6\,\mathrm{km}, 18\,\mathrm{h})$ also overflowed CHOLMOD's 32-bit interface (`attempts_log.md`, A5).

Sampler convergence at 10^6 observations. Nelder–Mead stopped far from the mode (best log ML 1,412,280 for the adapted kernel and 1,325,634 for the gp2Scale kernel), and the block Metropolis–Hastings chains were still climbing after 100 iterations (Fig. 5): the 60-iteration continuations raised the MAP log likelihood by a further 9,796 and 17,123, and the stage-3 chains gained 39,377 and 95,214 nats in 100 iterations without levelling off. With 10^6 observations the posterior is so concentrated that adapted random-walk proposals (cumulative acceptance 0.17–0.31 and 0.26–0.33) move by small fractions of the distance still to be travelled, so the chains behave as stochastic hill-climbers rather than as samplers of the posterior. We therefore use them only for MAP estimation; posterior uncertainty over θ is not reported, and at 45 s (stage 3: 141 s) per exact likelihood evaluation a gradient-based or Laplace-type treatment would be the natural next step.

Likelihood versus prediction for Lagrangian data. A drifter-memory term (Appendix D) raised the marginal likelihood of every pilot window (by 3,407–23,566 nats) and made held-out prediction markedly worse, because it lets same-drifter pairs explain the along-trajectory correlation that otherwise informs the shared field. Marginal likelihood is therefore not a safe guide to kernel design for Lagrangian data when the target is prediction at unseen instruments; we used it only within a fixed model family.

Calibration. Both exact GPs are over-dispersed where data are dense (90% intervals cover 0.993 of observations with ≥ 1000 training locations in the support) and under-dispersed at isolated locations (0.857), because a single amplitude function cannot express both the near-perfect interpolation inside dense clusters and the heavy-tailed errors of extrapolation; the aggregated 90% coverage (0.944) hides both. The two-scale model leaves this pattern essentially unchanged (0.990 and 0.863), and the stage-3 model lowers both coverages slightly (0.987 and 0.837), easing the over-dispersion inside clusters and deepening the under-dispersion at isolated locations.

6.3 What the exact posterior reveals about the flow

The near-inertial field is the dominant structured residual, and it is not submesoscale. The single most consequential fact the full-scale fit revealed about the LASER flow is visible in the training residuals alone (Fig. 7): after a 25-km/24-h background is removed, what remains is not primarily submesoscale noise but a clockwise oscillation at the inertial period that is coherent over at least 30 km, six times the support of the submesoscale kernels. Removing it with the harmonic background lowers the equal-time residual correlation at 20–30 km from 0.31 to 0.11 and the 6-hour rotary signal at that range from -0.34 to -0.10, and lowers the leave-one-drifter-out error of the background from 0.180 to 0.158 m s^{-1}, although the harmonic fit uses no information that the NW smoother does not have. In winter the Gulf's surface layer is repeatedly forced by atmospheric fronts, and this near-inertial response—energetic and phase-coherent over tens of kilometres, as in the slab picture of Pollard and Millard [1970]—sets a floor under any purely local covariance model of drifter velocities.

What remains is short-lived across drifters but long-lived along them. With the near-inertial part removed, the same-drifter residual correlation decays smoothly (0.80 at 3 h, 0.57 at 6 h, 0.30 at 12 h), whereas the best correlation between *different* drifters at a 3-hour lag is 0.45 even in a frame advected with the background flow. A large part of the residual variance travels with the water parcel or with the instrument, which is why the likelihood of trajectory data rewards drifter-specific memory and why only the shared part can be transferred to an unseen drifter (Sec. 6.2).

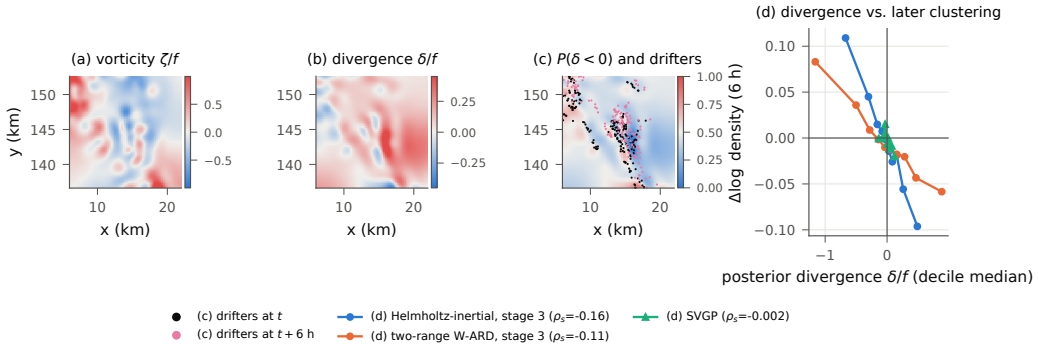


Figure 9: Kinematics read off the exact posterior of the near-inertial-covariance model (stage 3) on a 16-km box around the densest drifter cluster on 9 February 2016, 12:00 ($t = 492$ h; 0.25-km grid). (a) Posterior-mean vorticity and (b) divergence in units of f; (c) posterior probability of convergence with drifter positions at t (black) and $t + 6$ h (pink). (d) Mean change of $\log(1 + n)$, n the number of other training drifters within 1 km, over the next 6 h, per decile of posterior divergence at 29,637 training locations, for the two stage-3 kernels and SVGP (harmonic-residual baseline): convergence is followed by clustering under both exact GPs, most strongly under the adapted kernel, and not under SVGP.

Kinematics that only the exact posterior resolves. The exact posterior resolves submesoscale kinematics. At held-out locations the posterior-mean vorticity of the near-inertial-covariance model has an RMS of $1.39f$—a Rossby number of order one, as expected for submesoscale fronts [McWilliams, 2016, D'Asaro et al., 2018]—and the divergence $0.37f$, against $0.39f$ and $0.08f$ for SVGP, whose inducing-point posterior is smooth on scales of tens of kilometres (Fig. 9a–c). Pointwise the fields remain uncertain: $|\delta|$ exceeds two posterior standard deviations at 3.6% of the held-out locations and $|\zeta|$ at 12.6% (median posterior standard deviation of the divergence $0.54f$ against a prior value of $0.65f$). Collectively they carry information about where floating material accumulates. Across 29,637 training locations of 858 drifters, the posterior divergence is associated with the change of the local drifter density over the following six hours (Spearman $\rho_s = -0.157$, $p = 2 \times 10^{-162}$, drifter-bootstrap interval $[-0.170, -0.143]$), whereas SVGP's divergence on the same locations carries no such signal ($\rho_s = -0.002$, $[-0.016, +0.013]$; Fig. 9d). Every exact GP shows the association, but its strength depends on the covariance. The two kernels are indistinguishable in stages 1 and 2 (paired differences -0.005 $[-0.012, +0.002]$ and $+0.003$ $[-0.001, +0.008]$), whereas in stage 3 the adapted kernel's association is stronger than that of the same-support comparator ($\rho_s = -0.114$; difference -0.043 $[-0.052, -0.033]$), although its divergence field has the smaller RMS ($0.37f$ against $0.63f$). Long support without rotation strengthened the comparator's association only slightly over its two-scale version (-0.020 $[-0.025, -0.014]$): the divergence field is most informative about where drifters gather when the near-inertial motion is carried by the covariance, with its rotation. This is an association, not a forecast: the posterior at time t uses observations up to 24 h later. The covariance's sparsity pattern encodes the same physics: the 10% densest rows of $\mathbf{K}$ belong to locations where drifters have converged and sit in convergent flow (mean posterior divergence $-0.16f$, against $+0.01f$ for the sparser half of the rows; these rows hold 54% of all non-zeros of the 5-km pattern and 37% of the stage-3 pattern).

6.4 Limitations

The evidence has several limits. *Protocol.* Only stage 1 was pre-registered in full; stages 2 and 3 were designed after the stage-1 test errors had been seen, from training-only diagnostics and with every choice fixed before their own test evaluation, but their test results are not a pristine held-out confirmation. All stages share one frozen split of 48 drifters; the drifter bootstrap captures variability across drifters, not across splits or deployments. *Inference.* The hyperparameter chains did not converge, so we report plug-in predictions at the chain maximum without hyperparameter uncertainty; the background is a plug-in estimate whose own uncertainty enters only through the amplitude $a(z)$. *Model.* Hyperparameters are stationary over 100 days and over regimes from dense deployments to isolated drifters in the Loop Current; the contrast between pilot and full-scale fits (Sec. 6.1) suggests

that a non-stationary version, for instance with the amplitude and ranges of each term tied to local data support as in the gp2Scale non-stationary kernels, is the natural next step. Drifters sample the surface water preferentially where it converges, so the data are not a design independent of the field, which biases any spatial average (including our RMS kinematics) towards fronts. Drogued drifter velocities include slip and windage, part of the drifter-specific residual that no shared covariance can transfer to another drifter. *Evaluation.* The kinematic association with clustering is not a forecast, and the MNIST reproduction missed its pre-registered tolerance in the favourable direction for reasons we could not resolve. *Compute.* All runs used one node with 32 cores per job; several jobs shared the node, so wall-clock times include some contention.

7 Conclusion

Exact Gaussian-process regression on the full LASER drifter experiment—1,079,888 scalar velocity observations, a sparse covariance with up to 5.6×10^9 factor entries, exact log-determinants and solves, 32 CPU cores—is practical, and it beats a carefully tuned SVGP in RMSE, CRPS and log predictive density overall, and in RMSE and CRPS in every regime that has training data within the 5-km support, while matching it at isolated locations; SVGP's intervals are better calibrated only below the 95% level. The kernel that wins, however, is not the one we pre-registered. A compactly supported Helmholtz–inertial kernel confined to the 5-km support we had budgeted tied the gp2Scale Wendland kernel at full scale, because the dominant structured signal of the winter Gulf surface layer—near-inertial motion coherent over tens of kilometres—lay outside every kernel's reach, and three quarters of the error accrued where no training drifter was within that support. The fix is a matter of where each scale of motion lives: in a local harmonic mean (a cheap and robust improvement for any kernel) or, better, in the exact covariance itself, through a rotary term whose 30-km, 12-h support turned out to be affordable because the Cholesky factor of clustered drifter data grows only in proportion to its sparsity pattern. Under that support the rotary structure, which no kernel of the gp2Scale family can express, is decisive: the adapted kernel beats a two-range Wendland-ARD kernel with identical support, mean and protocol by 3.5% in RMSE and 7.3% in CRPS, and its learned frequency (chain median $1.03f$) lies at the inertial peak of the drifters' rotary spectrum ($1.025f$). Its posterior divergence is also more strongly associated with where drifters subsequently gather than the comparator's, while SVGP's carries no such signal. The kinematic (Helmholtz) part of the kernel, by contrast, received only a few per cent of the variance at full scale.

Three lessons extend beyond this data set. First, for exact GPs with compact support the binding constraint is not the factorisation but the scale separation of the process: diagnosing which correlations lie beyond the support should precede kernel design, and an exact covariance can afford far longer supports than its hyperparameter search can. Second, kernel rankings obtained on small windows did not survive at full scale in our experiments, so small-window pilot studies are an unreliable guide to the full-scale ranking of kernels. Third, for Lagrangian data the marginal likelihood rewards drifter-specific structure that cannot be transferred to an unseen instrument; held-out prediction, not likelihood, must arbitrate such additions. The natural next steps are non-stationary versions of the near-inertial and Helmholtz terms tied to local data support, posterior sampling of hyperparameters with gradient information, and forecasting experiments that use only past observations.

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A Positive definiteness and closed forms

Helmholtz terms. Let $k(\mathbf{s}, t; \mathbf{s}', t') = \kappa(\mathbf{s} - \mathbf{s}', t - t')$ with $\kappa(\Delta\mathbf{s}, \Delta t) = \frac{\sigma^2\rho^2}{22} w(q)$ and w the Wendland function $\psi_{3,3}$, which is positive definite on $\mathbb{R}^3$ and C^6. Then k is a valid covariance on $\mathbb{R}^2 \times \mathbb{R}$ and the process ψ with covariance k is twice mean-square differentiable, so $(\partial_x\psi, \partial_y\psi)$ is a Gaussian process with covariance $-\nabla\nabla^\top\kappa$; applying the rotation $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ gives the covariance $-J\nabla\nabla^\top\kappa J^\top$ of $\nabla^\perp\psi$. Both are covariances of explicit processes, hence positive semi-definite. For $q > 0$, $\partial_a\partial_b\kappa = \frac{\sigma^2}{22}[(w'' - w'/q)\, b_a b_b/q^2 + (w'/q)\, \delta_{ab}]$ (with $\mathbf{b} = \Delta\mathbf{s}/\rho$ and derivatives taken in units of ρ), and the polynomial identities $w'(q) = -22q(1-q)^7(16q^2 + 7q + 1)$ and $w''(q) - w'(q)/q = 528\, q^2(1+6q)(1-q)^6$ give Eq. (3); both expressions extend continuously to $q = 0$ with $f_0(0) = 1$, $f_1(0) = 24$. The support of each term is the ellipsoid $q < 1$.

Rotary term. With a, b independent and $\mathrm{Cov}(a(z), a(z')) = \mathrm{Cov}(b(z), b(z')) = \sigma_\iota^2 w(q_\iota)$, the vector $\mathbf{g}(z) = Q(\omega t)(a(z), b(z))^\top$ has $\mathrm{Cov}(\mathbf{g}(z), \mathbf{g}(z')) = \sigma_\iota^2 w(q_\iota)\, Q(\omega t) Q(\omega t')^\top = \sigma_\iota^2 w(q_\iota)\, Q(\omega(t - t'))$, which is Eq. (6).

Amplitude and mean. $a(z)a(z')\mathbf{K}(z, z')$ is positive semi-definite for any function a (Theorem 1 of Noack et al., 2025). The background $\mathbf{m}$ is a deterministic function of the training data and enters only as the prior mean.

Numerical checks. On random space–time point sets the assembled sparse matrices agree with dense references to machine precision, the smallest eigenvalues are positive, the posterior covariances of div under $\mathbf{K}_\psi$ and of curl under $\mathbf{K}_\phi$ vanish to $\sim 10^{-11}$, and the log marginal likelihood of our pipeline equals that of fvGP's own gp2Scale implementation to a relative error of 6.2×10^{-15} (`logs/tests/`).

B Divergence and vorticity posteriors

For a query point z we form the centred stencil $z \pm h\mathbf{e}_x$, $z \pm h\mathbf{e}_y$ with $h = 50\,\mathrm{m}$, so that $\mathrm{div}\,\mathbf{u}(z) \approx \mathbf{c}_{\mathrm{div}}^\top \mathbf{u}_{\mathrm{st}}$ and $\mathrm{curl}\,\mathbf{u}(z) \approx \mathbf{c}_{\mathrm{curl}}^\top \mathbf{u}_{\mathrm{st}}$ with fixed weights $\mathbf{c}$ on the eight stencil velocities. Since the posterior is Gaussian, $\mathbb{E}[\mathbf{c}^\top\mathbf{u}_{\mathrm{st}}] = \mathbf{c}^\top\boldsymbol{\mu}_{\mathrm{st}}$ and $\mathrm{Var}[\mathbf{c}^\top\mathbf{u}_{\mathrm{st}}] = \mathbf{c}^\top\mathbf{K}_{\mathrm{st,st}}\mathbf{c} - \|\mathbf{L}^{-1}\mathbf{P}\mathbf{K}_{X,\mathrm{st}}\mathbf{c}\|^2$, one forward solve with the stored Cholesky factor per query point and functional. Under a stream-function-only kernel the posterior divergence is 5×10^{-4} of the vorticity (pure stencil error), and the stencil variances agree with dense references to 10^{-11} relative error.

C Data provenance and protocol timeline

The LASER release (GRIIDC R4.x265.237:0001, doi:10.7266/N7W0940J) was downloaded on 2026-10-04 into the workspace; its SHA-256 checksum, the record counts after each filter and the frozen list of held-out drifters are in `logs/data_prep/data_prep_log.json`, produced by `code/scripts/prep_laser.py`. The split uses `numpy.random.default_rng(20160121)` at the level of physical drifters (drogued L/M segments of one drifter stay together). The pre-registration was committed before the first full-scale run (commit `ebbf028`); Addendum A (resumable execution, late pilot result) and Addendum B (bound asymmetry, budget-matched continuation, decided before any test prediction) followed during the stage-1 chains, Addendum C (two-scale model, commit `0a3391b`) after the stage-1 test evaluation and before any stage-2 run, Addendum D (free rotary frequency and its selection rule, commit `f404816`) during the stage-2 chains and before any stage-2 test prediction, and Addendum E (near-inertial covariance and its comparator, commit `3da888d`) before any stage-2 test result was read and before any stage-3 run (`method_plan.md`, `git log`). Every attempt that failed or was abandoned is recorded in `attempts_log.md`.

D Pilot study and abandoned kernel extensions

Table 2 lists the pilot fits that informed the pre-registration: Nelder–Mead MAP fits on 72-hour windows of training drifters, validated on 10% of the window's training drifters. The Helmholtz–inertial kernel won every pilot comparison by a wide margin in marginal likelihood and, with one exception, in validation error, which is why its full-scale tie was a genuine surprise.

Table 2: Pre-registration pilots (training drifters only; validation drifters held out within the window).

window	kernel	mean	n_{train}	log ML	RMSE	CRPS	cov. 90%
1–4 Mar	Wendland-ARD (gp2Scale)	constant	22,837	26,133	0.1700	0.0807	0.961
1–4 Mar	Helmholtz only	constant	22,837	39,672	0.1661	0.0765	0.966
1–4 Mar	Helmholtz–inertial	constant	22,837	43,664	0.1511	0.0710	0.988
1–4 Mar	Wendland-ARD (gp2Scale)	NW	22,837	48,584	0.1059	0.0494	0.972
1–4 Mar	Helmholtz–inertial	NW	22,837	57,666	0.1029	0.0469	0.978
1–4 Mar	Wendland-ARD (gp2Scale)	harmonic	22,837	56,271	0.0967	0.0462	0.984
1–4 Mar	Helmholtz–inertial	harmonic	22,837	60,968	0.0968	0.0442	0.978
8–11 Feb	Wendland-ARD (gp2Scale)	constant	41,444	96,009	0.1035	0.0440	0.957
8–11 Feb	Helmholtz only	constant	41,444	104,953	0.1118	0.0440	0.964
8–11 Feb	Helmholtz–inertial	constant	41,444	126,095	0.0829	0.0334	0.972
8–11 Feb	Wendland-ARD (gp2Scale)	NW	41,444	123,081	0.0778	0.0329	0.963
8–11 Feb	Helmholtz–inertial	NW	41,444	135,158	0.0732	0.0303	0.958
8–11 Feb	Wendland-ARD (gp2Scale)	harmonic	41,444	128,475	0.0710	0.0307	0.954
8–11 Feb	Helmholtz–inertial	harmonic	41,444	139,497	0.0672	0.0284	0.954

Drifter-memory (Lagrangian) term. Motivated by the along-trajectory correlations (Fig. 8), we added to either kernel a same-drifter term—the gp2Scale delta kernel with groups equal to drifters, multiplied by a compact temporal kernel—and a learned scale on the reported velocity errors (`wendland_ard_lag`, `helmholtz_inertial_lag`). On the dispersed pilot window the log marginal likelihood rose from 48,584 to 72,150 (Wendland-ARD) and from 57,666 to 70,290 (Helmholtz–inertial), but validation RMSE rose from 0.1059 to 0.1646 and from 0.1029 to 0.1468 m s^{-1}: the memory term absorbs the variance of the shared field (e.g. σ_ι fell from 0.144 to 0.040 m s^{-1}), so an unseen drifter is predicted almost from the mean alone. For Lagrangian data the marginal likelihood is dominated by along-trajectory autocovariance, and a channel that only same-drifter pairs can use wins it at the expense of the field that held-out prediction needs. The same happens in the dense deployment window, where cross-drifter information is plentiful: log likelihood 123,081→138,357 and validation RMSE 0.0778→0.1315 m s^{-1} for the Wendland-ARD kernel, and 135,158→138,565 and 0.0732→0.1121 m s^{-1} for ours. We therefore did not use the memory term at full scale.

Flow-following support. Shearing the support along the background flow (pairs selected by $|\Delta\mathbf{s} - \bar{\mathbf{m}}\Delta t|$ instead of $|\Delta\mathbf{s}|$) raised the best lagged cross-drifter residual correlation only from 0.49 to

0.59 at a 3-hour lag, and in both frames the correlation still peaked beyond the support, because drifters are displaced by the residual velocity itself (`logs/analysis/advection_diagnostic*.json`). We did not implement it.

E Secondary full-scale runs

Table 3: Secondary full-scale runs on the same test set: the budget-matched continuations of both stage-1 chains (Addendum B) and the stage-2 adapted kernel with the pre-registered lower bound on ω (Addendum D; the free-ω variant had the higher likelihood and is the stage-2 row of Table 1).

run	kernel	mean	log ML (MAP)	RMSE	CRPS	cov. 90%	NLPD
continuation of stage 1 (+60 iter.)	Helmholtz–inertial (ours)	NW	1,489,563	0.1459	0.0558	0.945	-1.221
continuation of stage 1 (+60 iter.)	Wendland-ARD (gp2Scale)	NW	1,517,569	0.1453	0.0554	0.944	-1.234
stage 2, $\omega \geq 0.17\,\mathrm{rad\,h^{-1}}$	Helmholtz–inertial (ours)	harmonic	1,585,338	0.1240	0.0498	0.945	-1.315

Likelihood probe of the stage-3 comparator. Midway through Nelder–Mead the two-range Wendland-ARD comparator (log likelihood 1,610,280) still had its long terms near their starting ranges (15 km, 9 h), whereas the adapted kernel's rotary term had already moved to 22 km and the 12-h cap. To check that the stage-3 gap was not an optimisation artefact, we evaluated the comparator on the training data with its long terms set by hand to the adapted kernel's ranges (22 km, 22 km, 12 h) and amplitudes of 0.10 and 0.13 $\mathrm{m\,s^{-1}}$, keeping the short terms fixed (`code/scratch/probe_ward2.py`, `logs/fullscale/ward2_long_probe.json`). The best probe reached 1,674,486 nats, 214,190 below the adapted kernel at the end of its Nelder–Mead stage (1,888,676); the comparator's own chain later climbed to 1,749,900. The protocol was not changed, and no test data were used.

F Coverage curves

Table 4 lists the empirical coverage of central predictive intervals on all 56,702 held-out scalar observations for every full-scale run, including the secondary runs of Table 3, and for the three SVGP configurations (mean over three seeds; the seed standard deviation is at most 0.004 at every level). These are the curves of Fig. 2.

Table 4: Empirical coverage of central predictive intervals at each nominal level (held-out drifters).

nominal level	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95	0.99
stage 3: Helmholtz–inertial, 30-km rotary term	.199	.370	.508	.617	.704	.772	.830	.881	.930	.954	.980
stage 3: two-range Wendland-ARD	.211	.392	.536	.644	.726	.792	.846	.894	.938	.960	.983
stage 2: Helmholtz–inertial	.209	.389	.528	.636	.723	.793	.852	.901	.944	.966	.985
stage 2: Wendland-ARD	.202	.381	.523	.632	.721	.793	.852	.900	.945	.967	.986
stage 1: Helmholtz–inertial	.230	.414	.550	.652	.732	.798	.854	.900	.944	.965	.984
stage 1: Wendland-ARD	.208	.387	.528	.636	.722	.792	.851	.900	.944	.966	.985
stage 1 continuation: Helmholtz–inertial	.225	.408	.545	.650	.731	.799	.855	.902	.945	.966	.984
stage 1 continuation: Wendland-ARD	.206	.385	.526	.635	.721	.790	.849	.899	.944	.965	.984
stage 2, $\omega \geq 0.17$: Helmholtz–inertial	.210	.391	.529	.638	.725	.796	.855	.903	.945	.967	.986
SVGP, harmonic residual	.139	.274	.398	.508	.608	.694	.772	.841	.903	.934	.968
SVGP, velocities	.139	.274	.399	.509	.608	.696	.773	.843	.908	.942	.975
SVGP, NW residual	.148	.286	.413	.525	.623	.707	.782	.848	.909	.939	.971

G Background selection

All bandwidths and shrinkage parameters were chosen by leave-one-drifter-out (LODO) error on training drifters, the protocol used for the pre-registered background (Table 5). "Isolated" locations

have no other-drifter training location inside the $(5\,\mathrm{km}, 24\,\mathrm{h})$ support (17.8% of training locations); "frontier" locations have NW weight $W < 10$. Short windows ($L = 15\,\mathrm{km}$, $T = 8\,\mathrm{h}$) minimise the overall LODO error because they act as a local interpolator, but they are worse at isolated locations, which are the only places where the background is not superseded by the GP.

Table 5: Leave-one-drifter-out RMSE ($\mathrm{m\,s^{-1}}$) of background estimators on training drifters.

background	L (km)	T (h)	κ	LODO RMSE	isolated	frontier
Nadaraya–Watson	25	24	0	0.1797	0.2388	0.3309
Nadaraya–Watson	25	24	1	0.1784	0.2330	0.3124
Nadaraya–Watson	25	24	3	0.1788	0.2340	0.3160
harmonic (Eq. 8)	25	24	0	0.1598	0.2282	0.3313
harmonic (Eq. 8)	25	24	1	0.1582	0.2216	0.3111
harmonic (Eq. 8)	25	24	3	0.1586	0.2227	0.3145
harmonic (Eq. 8)	40	24	1	0.1670	0.2257	0.3080
harmonic (Eq. 8)	40	24	3	0.1668	0.2243	0.3033
harmonic (Eq. 8)	15	8	0	0.1399	0.2394	–

H Baseline tuning

Table 6: SVGP configurations on the validation split (10% of training drifters, seed 7). Two independent latent GPs, natural gradients for the variational parameters (learning rate 0.1), Adam (0.01) for hyperparameters, mini-batches of 4,096 locations.

kernel	target	M	epochs	RMSE	CRPS	cov. 90%	time (min)
Matérn-3/2	NW residual	1000	15	0.1581	0.0825	0.905	8
Matérn-3/2	harmonic residual	2000	15	0.1346	0.0680	0.895	35
Matérn-3/2	harmonic residual	3000	12	0.1363	0.0681	0.890	63
Matérn-3/2	velocities	1000	10	0.1523	0.0806	0.906	5
Matérn-3/2	velocities	1000	15	0.1523	0.0805	0.902	8
Matérn-3/2	velocities	2000	15	0.1447	0.0751	0.899	30
RBF	velocities	1000	15	0.1613	0.0860	0.903	8

I Cost of exactness versus support radius

Table 7 counts the location pairs and stored entries of the $1{,}079{,}888 \times 1{,}079{,}888$ pattern for space–time supports $(\rho, \rho, 24\,\mathrm{h})$ and, where affordable, the entries and wall-clock of one supernodal Cholesky factorisation of the Wendland-ARD covariance at full support. Pairs grow far more slowly than ρ^2 because most data sit in a few dense clusters that are already fully connected at small radii, and the factor grows in proportion to the pattern (about 1.36–1.39 factor entries per stored entry at every radius we factorised), so even a 30-km support would leave the exact computation within reach of a single node; what grows prohibitively is the cost of the hyperparameter search, which needs hundreds of factorisations (Sec. 6.2).

Table 7: Pattern and Cholesky-factor size versus support (training set; supports given as (spatial range in km, temporal range in h), unions of two ellipsoids below the line). Factorisation times are single measurements on 32 cores while other jobs shared the node.

support (km, h)	pairs	stored entries	non-zero	factor entries	factor (GB)	time (s)
(3, 24)	8.76×10^7	6.99×10^8	0.060%	9.71×10^8	8	29
(4, 24)	1.24×10^8	9.93×10^8	0.085%	1.35×10^9	11	47
(5, 24)	1.62×10^8	1.29×10^9	0.111%	1.77×10^9	14	67
(6, 24)	1.99×10^8	1.59×10^9	0.137%	2.20×10^9	18	84
(7, 24)	2.35×10^8	1.88×10^9	0.161%	2.60×10^9	21	87
(8, 24)	2.69×10^8	2.15×10^9	0.184%	2.98×10^9	24	98
(10, 24)	3.33×10^8	2.66×10^9	0.228%	–	–	–
(12, 24)	3.92×10^8	3.13×10^9	0.269%	–	–	–
(15, 24)	4.72×10^8	3.78×10^9	0.324%	–	–	–
(20, 24)	5.84×10^8	4.67×10^9	0.400%	–	–	–
(30, 24)	7.50×10^8	6.00×10^9	0.514%	–	–	–
(5, 24) ∪ (30, 6)	2.82×10^8	2.25×10^9	0.193%	4.13×10^9	33	106
(5, 24) ∪ (20, 12)	3.54×10^8	2.83×10^9	0.243%	4.43×10^9	35	111
(5, 24) ∪ (30, 12)	4.31×10^8	3.44×10^9	0.295%	5.63×10^9	45	152

J MNIST reproduction

Table 8: MNIST $\Pr(y = 5)$ regression, random 60,000/10,000 splits. The gp2Scale rows use the provided fvGP code with the kernel $k_{\mathcal{W}} k_d$ (ℓ_1 Wendland times the nearest-neighbour delta kernel, $k = 20$) and 120 BMH iterations; Brier scores are computed on predictions clipped to $[0, 1]$ ("clipped") and unclipped ("raw").

method	seed	Brier (clipped)	Brier (raw)	accuracy
fvGP gp2Scale, $k_{\mathcal{W}} k_d$ (ours)	0	0.0099	0.0114	0.9905
fvGP gp2Scale, $k_{\mathcal{W}} k_d$ (ours)	1	0.0106	0.0120	0.9888
fvGP gp2Scale, $k_{\mathcal{W}} k_d$ (ours)	2	0.0102	0.0118	0.9904
SVGP, 500 inducing points (ours)	0	0.0142	0.0150	0.9907
SVGP, 500 inducing points (ours)	1	0.0130	0.0144	0.9881
SVGP, 500 inducing points (ours)	2	0.0132	0.0145	0.9894
gp2Scale (reported)	–	0.018 ± 0.002		–
SVGP (reported)	–	$0.033 \pm 2.8 \times 10^{-5}$		–